EasySeva

Government Services, Made Easier

A Citizen-First Web Portal — Human–Computer Interaction Design Report

A Project Report submitted for the course

Human–Computer Interaction

Submitted by:

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Live application: `https://easyseva-rho.vercel.app`

Source repository: `https://github.com/rZk1809/easyseva`

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1 Abstract

EasySeva is a responsive web portal that helps ordinary citizens *discover*, *apply for*, and *track* government services — with a Tamil Nadu service context — in their own language and at their own pace. The project was built as a Human–Computer Interaction (HCI) prototype whose primary users are first-time and low-digital-literacy citizens who often find official portals intimidating, jargon-heavy, and error-prone.

This report explains what the application does, why it is useful, and — in depth — how established HCI principles were deliberately applied on every screen. We analyse the interface against four complementary frameworks: Jakob Nielsen's ten usability heuristics, Ben Shneiderman's eight golden rules of interface design, Don Norman's principles of interaction design, and the WCAG POUR accessibility pillars. For each page of the application we provide a screenshot, describe its purpose, and enumerate the specific principles used, stating both *what the principle means* and *how it is realised in the interface*. The report closes with a heuristic-coverage matrix and a reflection on how good HCI directly improves the citizen's experience.

2 Introduction

2.1 What is EasySeva?

Applying for a government document — an income certificate, a scholarship, a ration card, an old-age pension — is a task almost every household faces. Yet the digital experience is typically poor: dense forms with thirty fields on one page, unexplained acronyms, English-only interfaces, no way to save progress, and status updates that read like internal department codes. For a first-time or less-confident user this is a wall.

EasySeva reimagines that experience around *how citizens think* rather than how departments are organised. It offers:

- A catalogue of **nine services** across six citizen-centred categories (Certificates, Benefits, Education, Family, Senior Citizen, and Document Updates).
- A **guided, step-by-step** application flow — eligibility check, document checklist, chunked form, review, and confirmation.
- **Passwordless OTP login**, **save-and-resume** drafts, and **application tracking** by reference number.
- A **trilingual** interface (English, Tamil, Hindi) and built-in **accessibility** controls (large text, high contrast, reduced motion).

2.2 Who is it for, and why is it useful?

The target users are citizens who are intimidated by bureaucracy: first-time applicants, elderly users, and people with limited English or limited digital experience. EasySeva is useful because it lowers every barrier that usually causes abandonment or error: it speaks plain language, it shows exactly what is needed before the user starts, it validates input as it is typed, it never loses the user's work, and it explains every status in human terms. The result is that a citizen can complete an application correctly *the first time*, with confidence and dignity.

2.3 How HCI enables a good interaction

Human–Computer Interaction is the study of designing systems around human abilities, limitations, and expectations. A government portal is exactly the kind of *high-stakes, low-frequency, high-anxiety* task where good HCI matters most: the user may use it once a year, cannot afford

to make a mistake, and has no training. By applying well-understood principles — visibility of status, error prevention, recognition over recall, user control, and universal accessibility — EasySeva turns an anxious, failure-prone chore into a calm, predictable, and forgiving experience. The rest of this report demonstrates, screen by screen, how each principle was applied.

3 HCI Principles and Frameworks Used

This section defines the frameworks referenced throughout the page-by-page analysis.

3.1 Nielsen's 10 Usability Heuristics

N1. **Visibility of system status** — keep users informed through timely, appropriate feedback.

N2. **Match between the system and the real world** — speak the users' language with familiar words and concepts.

N3. **User control and freedom** — provide clearly marked exits, undo, and the freedom to change course.

N4. **Consistency and standards** — follow platform and internal conventions so similar things look and behave similarly.

N5. **Error prevention** — design to prevent problems before they occur, not just to report them.

N6. **Recognition rather than recall** — make options, actions, and information visible so users need not remember things.

N7. **Flexibility and efficiency of use** — allow both novices and experts to work in the way that suits them.

N8. **Aesthetic and minimalist design** — show only what is relevant; remove clutter that competes for attention.

N9. **Help users recognise, diagnose, and recover from errors** — express errors in plain language with a constructive next step.

N10. **Help and documentation** — provide easy-to-search, task-focused help where it is needed.

3.2 Shneiderman's 8 Golden Rules

S1. Strive for consistency.

S2. Cater to universal usability (novices to experts, accessibility).

S3. Offer informative feedback.

S4. Design dialogs to yield closure.

S5. Prevent errors.

S6. Permit easy reversal of actions.

S7. Keep users in control (internal locus of control).

S8. Reduce short-term memory load.

3.3 Norman's Interaction Design Principles

D1. **Affordances and signifiers** — controls look like what they do and signal how to use them.

D2. **Feedback** — every action produces a perceivable result.

D3. **Mapping** — the relationship between controls and effects is natural (e.g., a progress bar maps to steps completed).

D4. **Constraints** — the design limits invalid actions.

D5. **Consistency** — similar operations use similar elements.

D6. **Conceptual model** — the interface helps users form a correct mental model of the process.

3.4 WCAG Accessibility Pillars (POUR)

Content must be **Perceivable**, **Operable**, **Understandable**, and **Robust**. EasySeva addresses POUR through colour-contrast-aware theming, full keyboard operability with visible focus, plain-language content, and semantic, assistive-technology-friendly markup.

4 Design System and Foundations

Before the page analysis, three cross-cutting foundations are worth noting because they support *consistency*, *universal usability*, and *accessibility* on every screen.

4.0.1 A consistent visual language (N4, S1, D5)

All screens are built from a small set of reusable primitives — Button, Card, Input, Alert, Dialog, Badge, Progress — driven by shared design tokens (a navy/blue/teal palette, consistent radius and spacing scales, and a single Inter typeface). This guarantees that a primary action, a status badge, or a form field looks and behaves identically everywhere, so learning one screen teaches the user every screen.

4.0.2 Built-in accessibility controls (S2, POUR)

Every page can be adapted by the user: three text sizes, a high-contrast theme, and a reduced-motion mode that respects the operating-system preference. A "skip to main content" link, visible keyboard focus rings, 44–56 px touch targets, and ARIA roles/labels are present throughout.

4.0.3 Trilingual interface (N2, N7)

The entire interface is available in English, Tamil, and Hindi. Language can be chosen on the landing page or switched at any time from the header; the choice is remembered across visits.

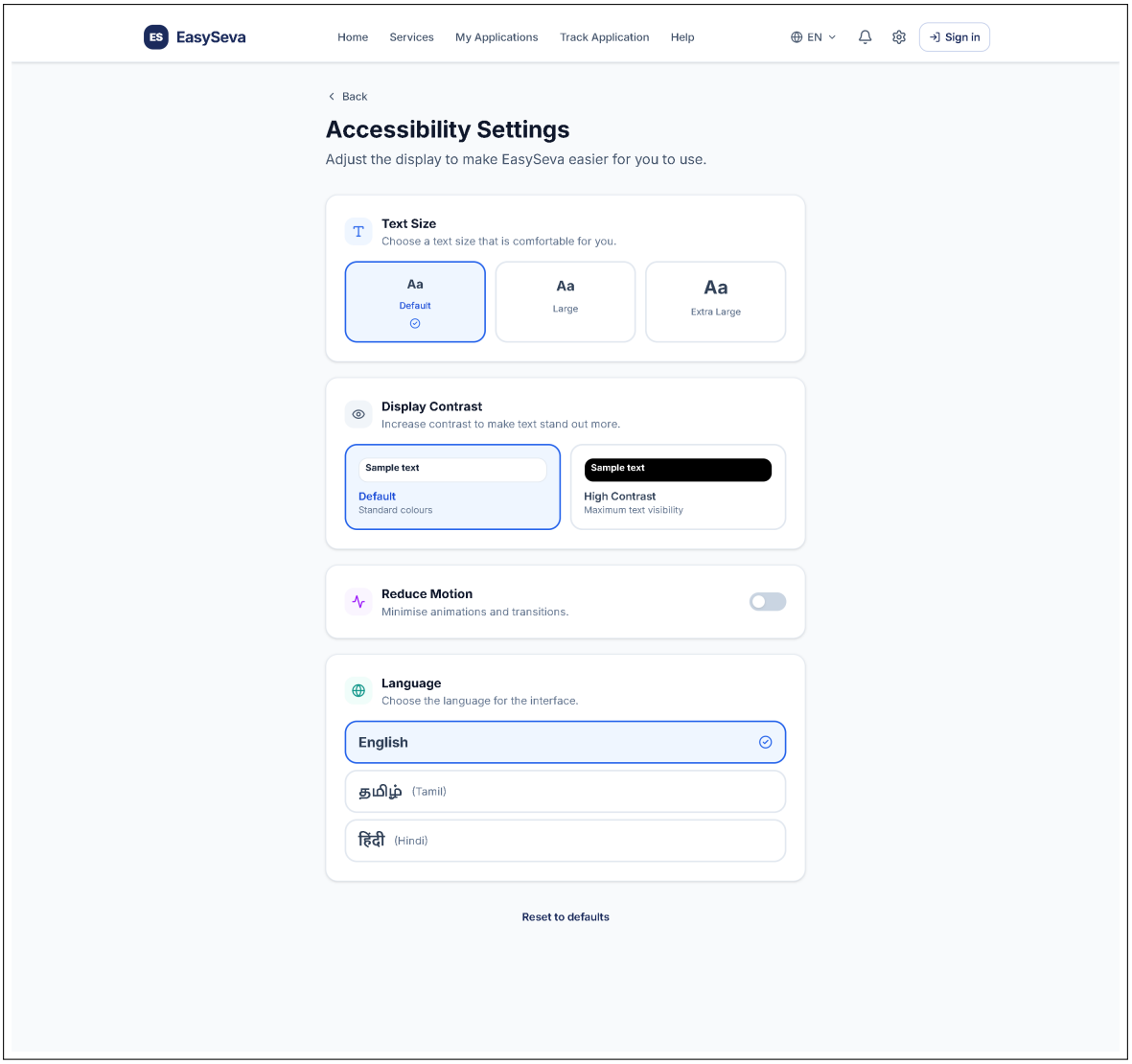


Figure 1. The Accessibility Settings page lets each user tune text size, contrast, motion, and language — an explicit realisation of Shneiderman's rule to "cater to universal usability."

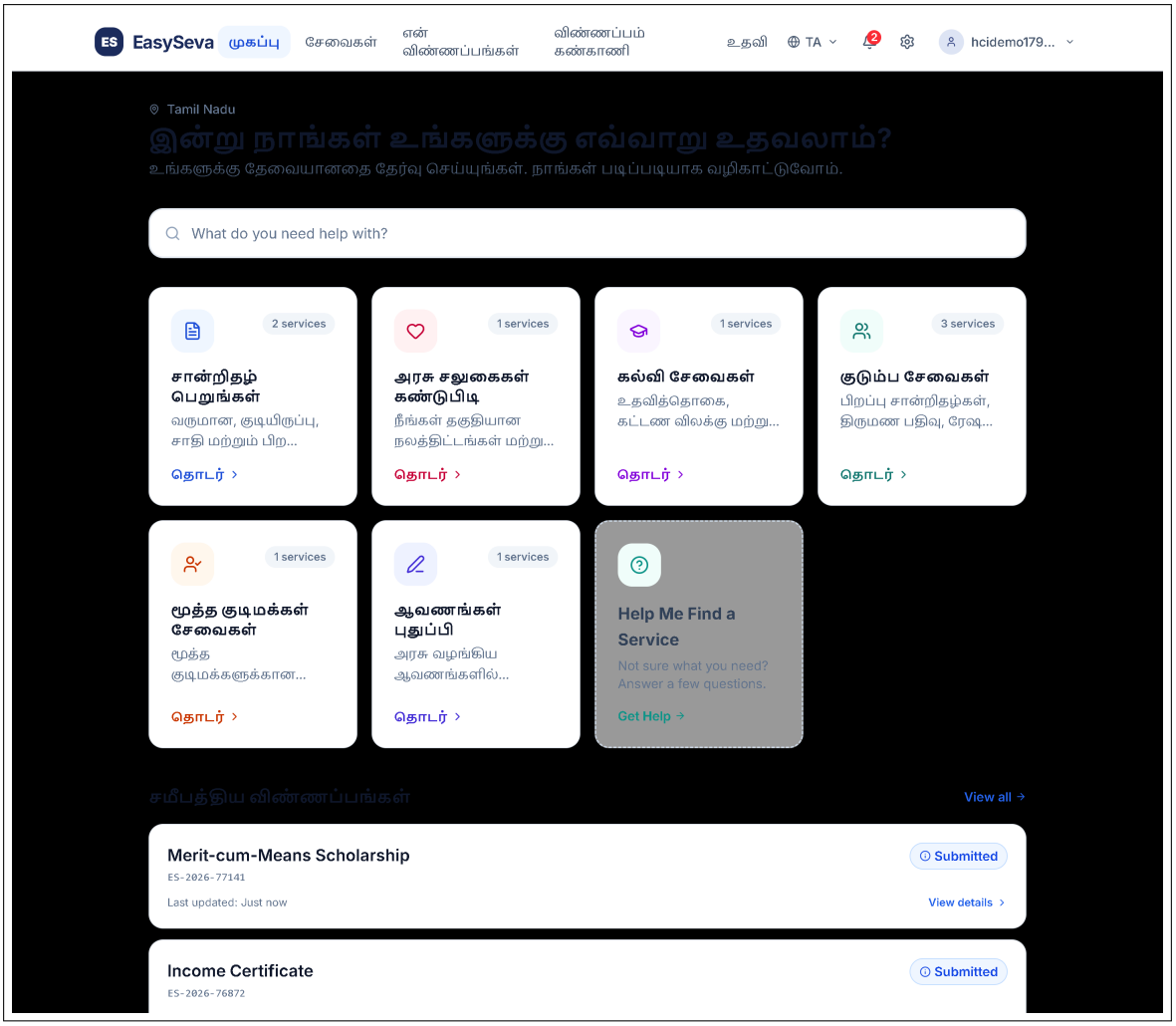


Figure 2. High-contrast mode (shown here together with the Tamil interface). The dark, high-contrast theme improves perceivability for low-vision users and demonstrates that the accessibility and language settings apply globally and persist across pages.

5 Page-by-Page HCI Analysis

For each screen we state its *purpose* and then enumerate the principles applied, giving both the meaning of the principle and the concrete way it is realised in the interface.

5.1 Landing Page

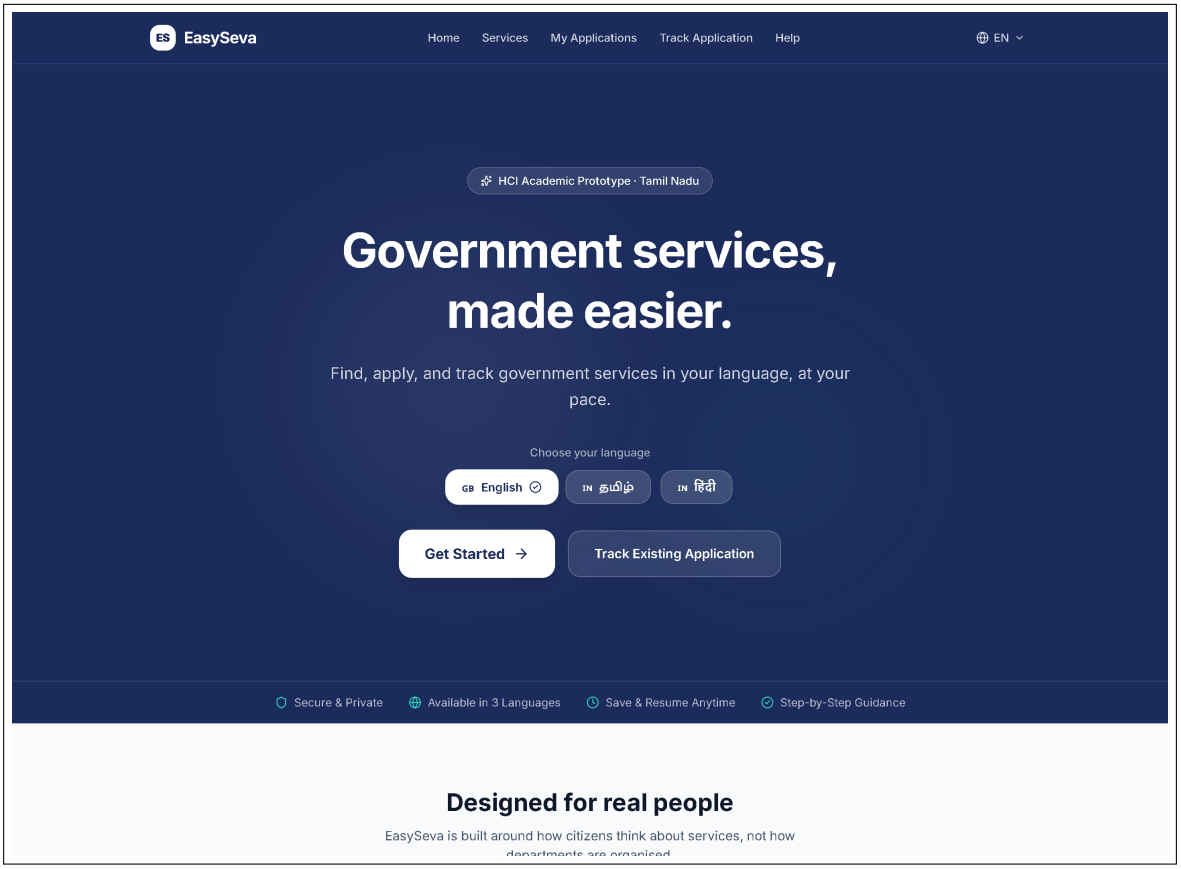


Figure 3. The landing page: a single clear value proposition, an up-front language choice, and two obvious next actions.

Purpose. Create trust in seconds and route the user to the one action they came for — start a service or track an existing one.

- **Aesthetic and minimalist design (N8).** *Show only what matters*
  A single headline, one sub-line, the language choice, and exactly two buttons. There is no clutter competing for the first-time user's attention.
- **Match with the real world (N2).** *Use the user's language*
  The promise "Government services, made easier" and "Find, apply, and track . . . in your language, at your pace" is written the way a citizen thinks, not in departmental terminology.
- **Flexibility and efficiency (N7).** *Serve diverse users*
  Language is chosen before anything else, so a Tamil- or Hindi-first user is never forced through an English wall (see Figure 4).
- **Recognition rather than recall (N6).** *Make choices visible*
  The trust strip (Secure & Private, 3 Languages, Save & Resume, Step-by-Step) surfaces the system's key capabilities instead of expecting the user to discover them.
- **Affordance and signifiers (D1).** *Controls look actionable*
  The solid "Get Started" button clearly affords the primary action; the secondary "Track" button is visually subordinate, mapping importance to appearance.

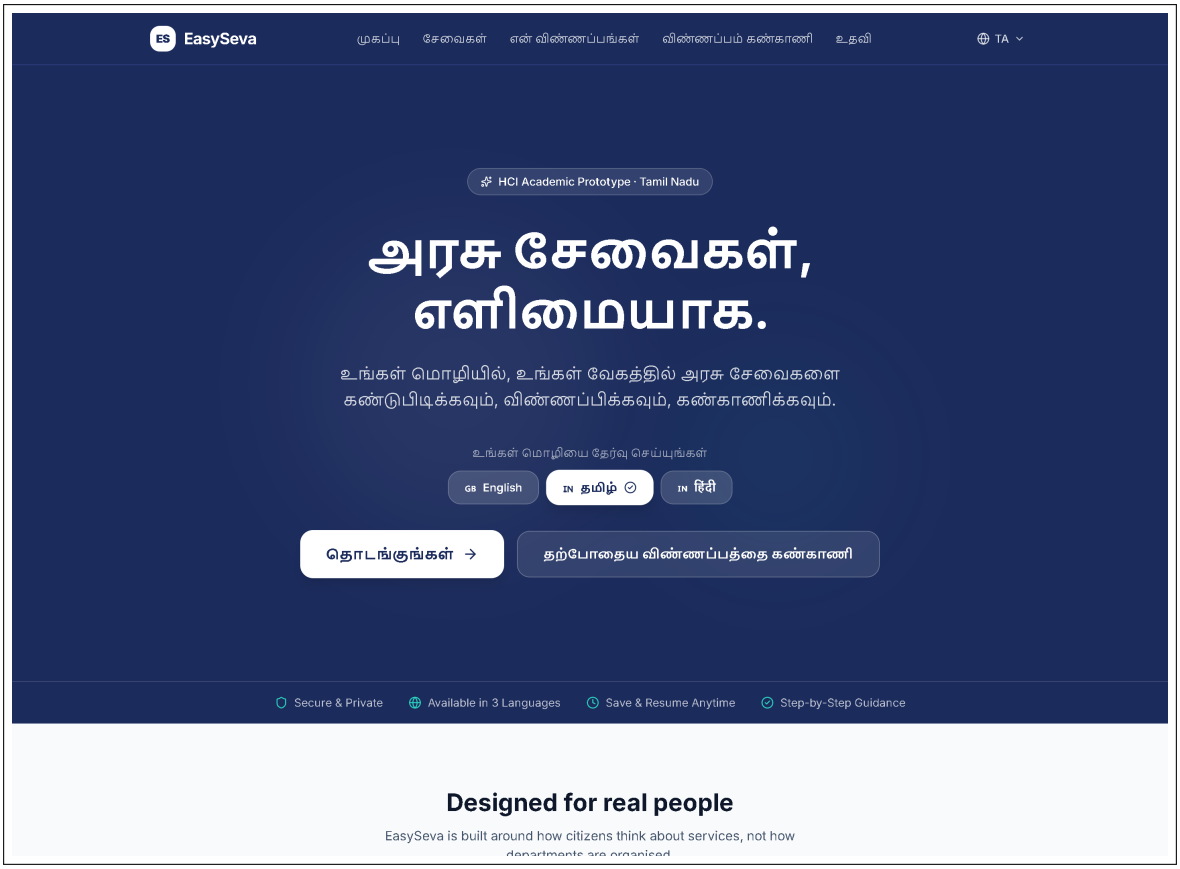


Figure 4. The same landing experience in Tamil. Choosing a language rewrites the entire interface, directly supporting "match with the real world" and "flexibility of use" for non-English speakers.

5.2 Home Dashboard

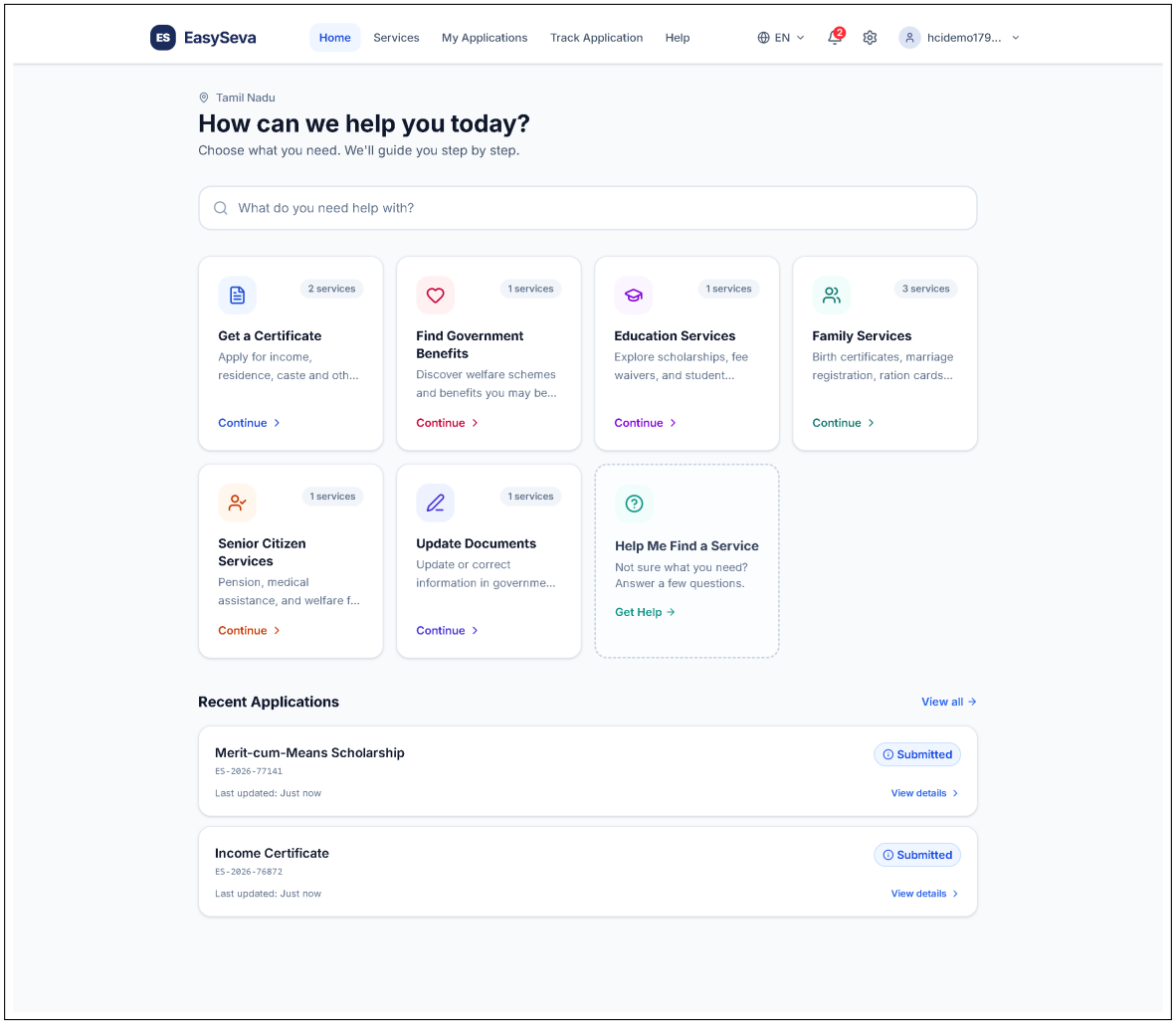


Figure 5. The home dashboard: task-based service categories, a natural- language search, a "Help me find a service" escape hatch, and the user's recent applications with live status.

Purpose. Give the signed-in citizen a single hub to begin a new task or resume an existing one.

- **Match with the real world (N2).** *Organise by user goals*
  Services are grouped as "Get a Certificate," "Find Government Benefits," "Family Services" — citizen intents, not government department names.
- **Recognition rather than recall (N6).** *Show, don't make them remember*
  Each category is a large card with an icon, title, and example services, so the user recognises the right path instead of recalling a menu.
- **Visibility of system status (N1, S3).** *Keep users informed*
  The "Recent Applications" section shows each application's reference number and current status (e.g., "Submitted") so the user always knows where things stand.
- **Flexibility and efficiency (N7).** *Multiple routes*
  Confident users can type a need into the search box; unsure users can click "Help Me Find a Service" to be guided by questions.

5.3 Service Catalogue and Category Pages

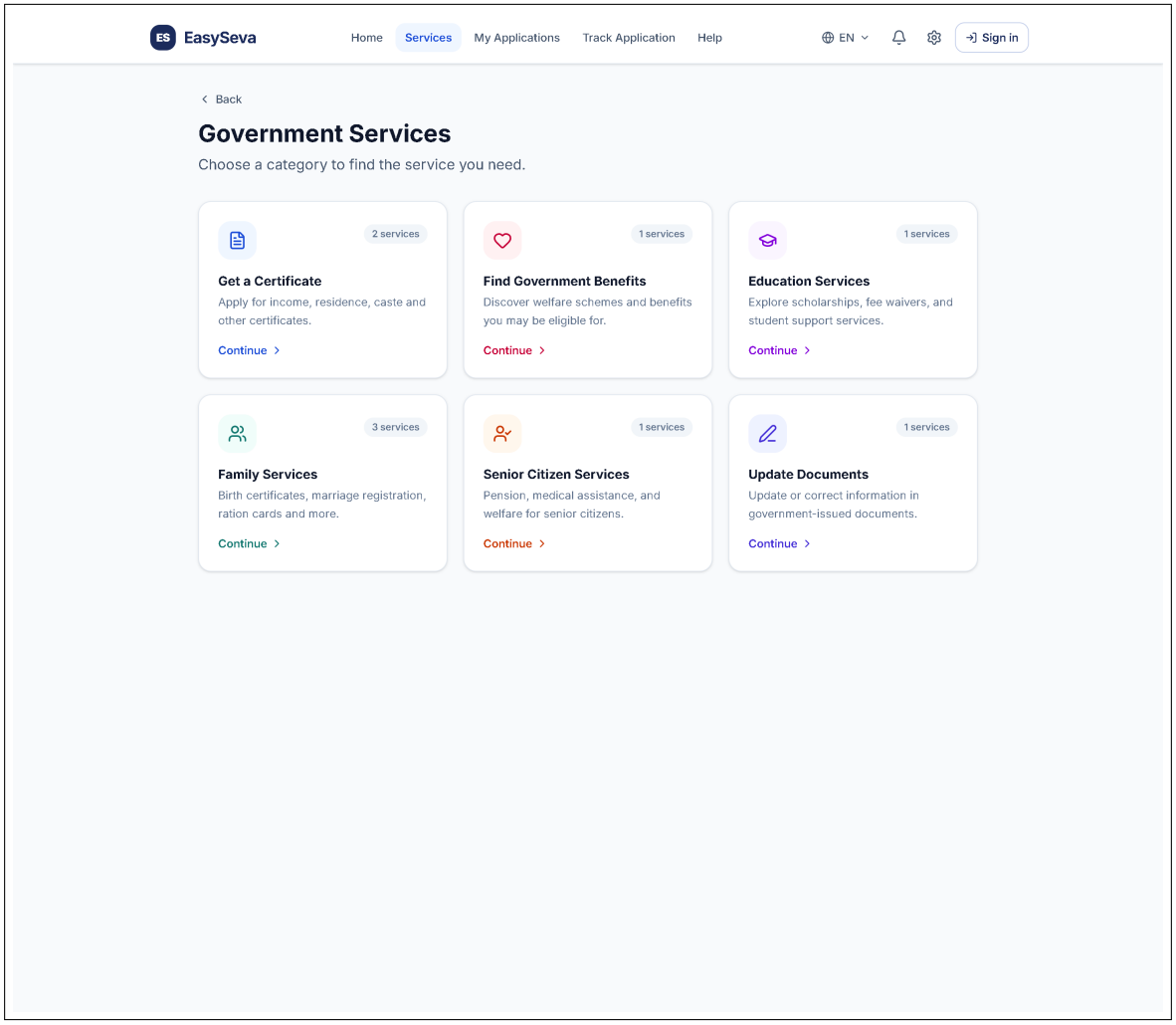


Figure 6. The full service catalogue, grouped into the six citizen- centred categories with an accurate count of services in each.

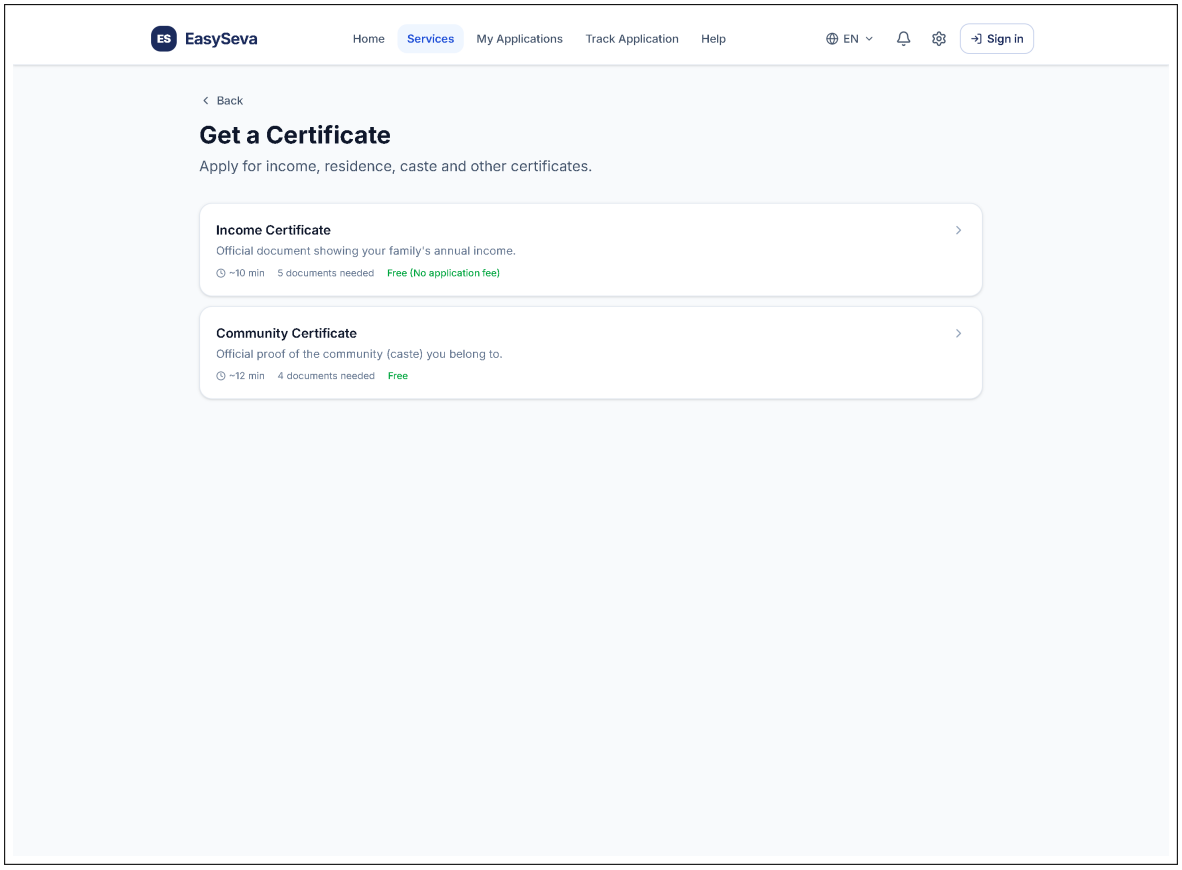


Figure 7. A category page (Certificates) listing the services it contains.

Purpose. Let users browse the full catalogue or drill into a category.

- **Consistency and standards (N4, S1).** *Similar things look similar*
  Every service and category is presented with the same card component, so browsing is predictable across the whole catalogue.
- **Recognition rather than recall (N6).** *Visible structure*
  Category icons and short descriptions let users recognise where a service lives.
- **Aesthetic and minimalist design (N8).** *Reduce overload*
  A large catalogue is chunked into six digestible groups rather than one long list.

5.4 Service Detail Page

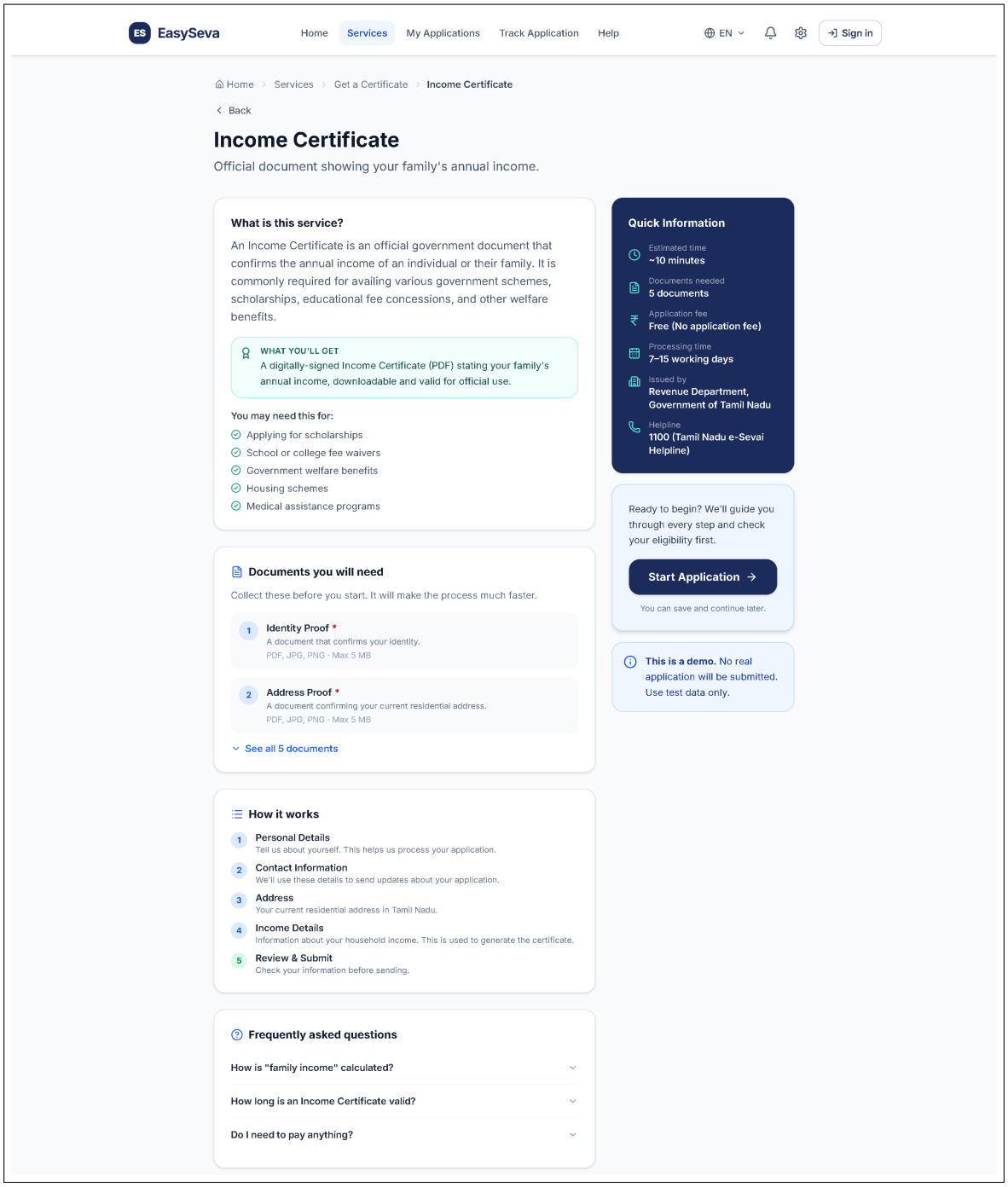


Figure 8. The service detail page for Income Certificate: breadcrumbs, a plain-language explanation, "What you'll get," a document checklist, a "How it works" overview, FAQs, and a Quick Information panel with time, fee, processing duration, issuing department, and helpline.

Purpose. Help the user decide, with confidence, whether and how to apply — *before* committing time.

- **Help and documentation (N10).** *Task-focused help in place*
  "What is this service?," "You may need this for," and a dedicated FAQ accordion answer the citizen's real questions at the moment of decision.
- **Visibility of system status (N1).** *Set expectations*

The Quick Information panel states estimated time, fee, processing duration, the issuing department, and a helpline, so there are no surprises later.

- **Recognition rather than recall (N6).** *Orientation*
  Breadcrumbs (Home / Services / Certificates / Income Certificate) always show where the user is and allow one-click return to any level.
- **Error prevention (N5, S5).** *Prepare before starting*
  The "Documents you will need" checklist lets users gather everything *before* they begin, preventing mid-form abandonment.
- **User control and freedom (N3, S7).** *No forced commitment*
  A reassuring "You can save and continue later" note and a clear single primary action ("Start Application") keep the user in control.

5.5 Eligibility Check

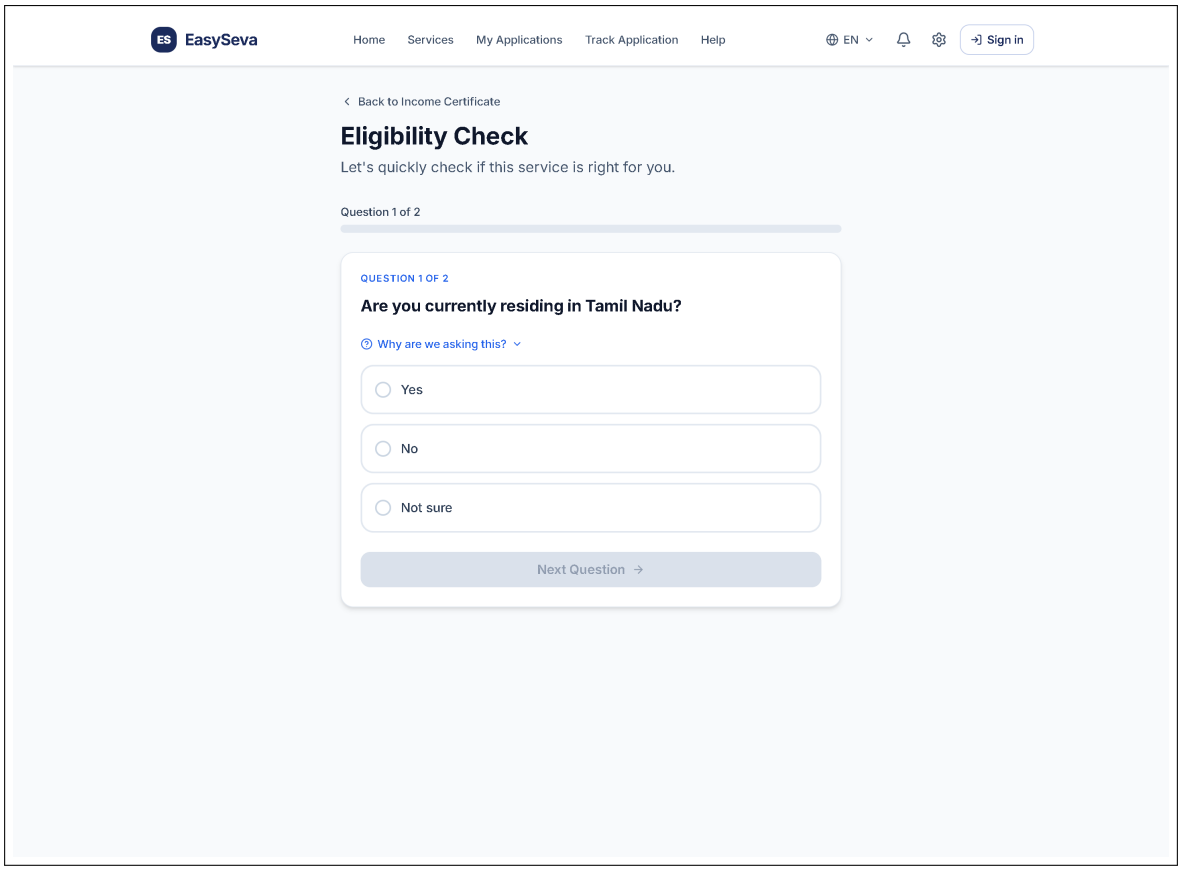


Figure 9. The eligibility check asks one plain question at a time, with an optional "Why are we asking this?" explanation and a progress bar.

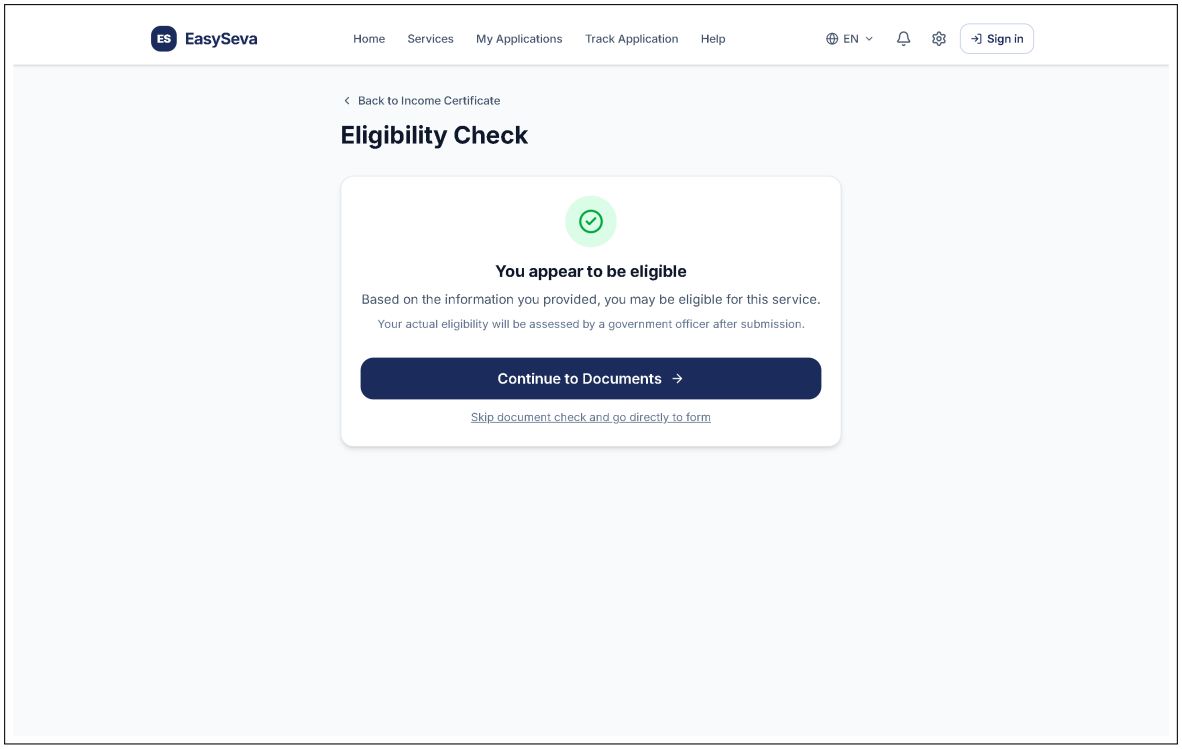


Figure 10. The result screen gives a clear verdict and — even when the user may not qualify — preserves their freedom to continue or to find a more suitable service.

Purpose. Prevent wasted effort by checking suitability up front, without discouraging the user.

- **Error prevention (N5, S5).** *Stop problems early*
  Checking eligibility before the form prevents the larger "filled everything, then rejected" failure.
- **Help and documentation (N10).** *Explain the ask*
  Each question offers a "Why are we asking this?" disclosure, building understanding and trust.
- **Visibility of system status (N1, D3).** *Progress mapping*
  A "Question 1 of 2" progress bar maps directly to how far the user has come.
- **User control and freedom (N3, S7).** *Respect autonomy*
  An ineligible result never blocks the user: they may "Continue Anyway" (an officer decides) or ask for help finding a different service.
- **Informative feedback (S3).** *Announce the outcome*
  The result is wrapped in an `aria-live` region so screen-reader users hear the verdict immediately.

5.6 Document Upload

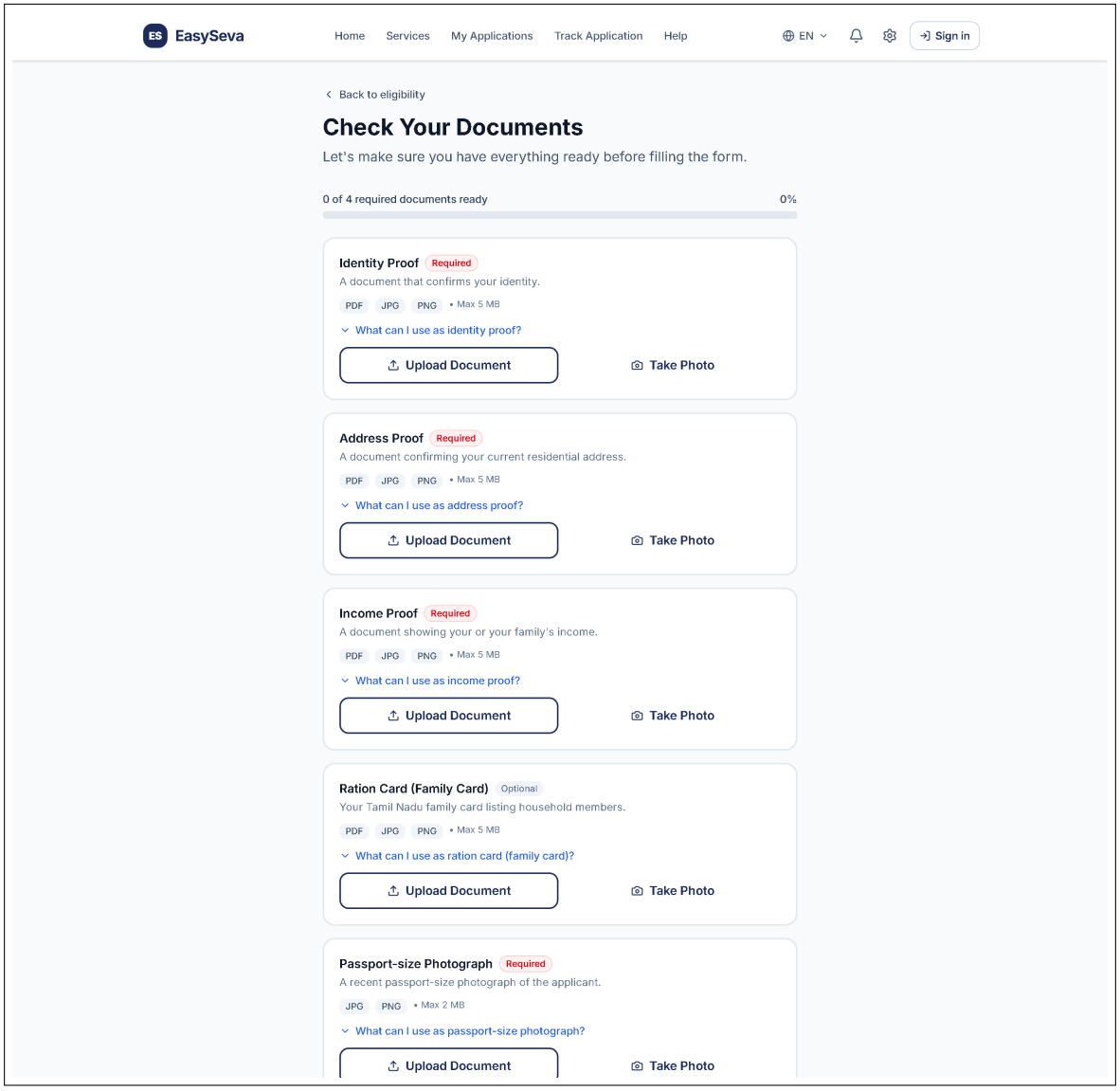


Figure 11. The document step: each requirement is a self-contained card with accepted formats, a size limit, examples of acceptable documents, and upload or "take photo" actions.

Purpose. Make uploading the right documents effortless and error-free.

- **Recognition rather than recall (N6).** *Show acceptable options*
  A "What can I use as . . . ?" disclosure lists concrete examples (Aadhaar, Voter ID, . . .) so the user need not recall what qualifies.
- **Error prevention (N5, S5).** *Validate before accepting*
  Only PDF/JPG/PNG within the size limit are accepted; invalid files are rejected with a clear reason before any submission.
- **Visibility of system status (N1, S3).** *Confirm success*
  An uploaded document turns green with a check and its file name and size, so the user sees it was accepted.
- **Flexibility and efficiency (N7).** *Fit the device*
  Both "Upload Document" and "Take Photo" are offered — the latter is ideal on a phone.

- **Constraints (D4).** *Guide valid input*
  The file picker is restricted to accepted file types, constraining the user toward a valid choice.

5.7 Application Form

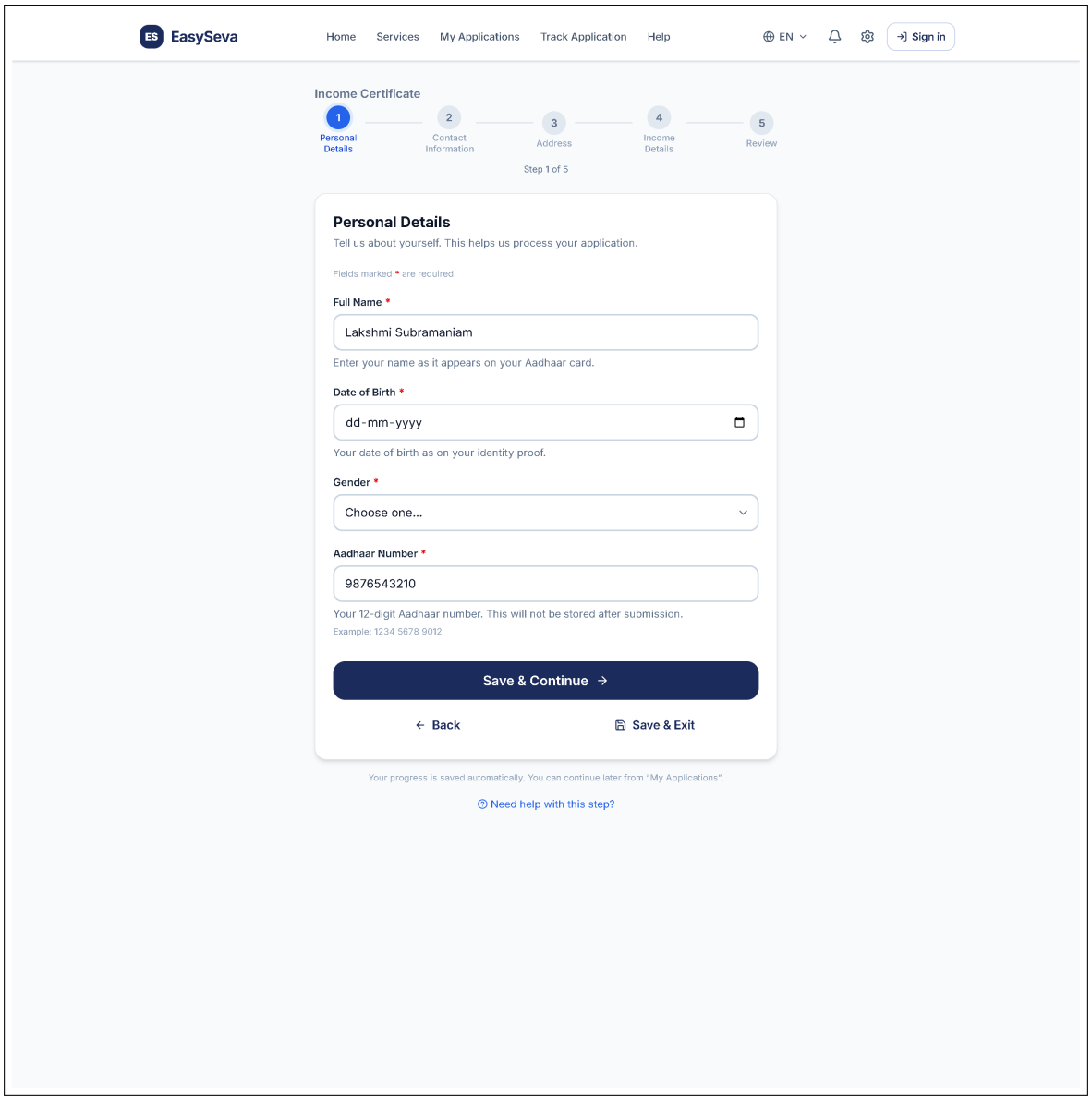


Figure 12. The form presents one small step at a time, with plain labels, helper text, examples, and inline validation — never a wall of thirty fields.

Purpose. Collect accurate information with the least possible cognitive load.

- **Reduce short-term memory load (S8, N6).** *Chunk the work*
  The form is split into small, themed steps (Personal, Contact, Address, Income) instead of one overwhelming page.
- **Error prevention (N5, S5).** *Catch mistakes as they happen*
  Fields validate inline (e.g., a 12-digit Aadhaar, a 10-digit mobile, a 6-digit PIN) with specific, friendly messages.

- **Match with the real world (N2).** *Human labels*
  Labels and helper text use everyday phrasing ("Estimated Annual Income of Household") with worked examples.
- **Visibility of system status (N1, D3).** *Show progress*
  A step indicator shows how many steps remain, mapping effort to completion.
- **User control and freedom (N3, S6).** *Never lose work*
  "Save & Finish Later" persists a draft at any point; users can leave and return safely.

5.8 Review Before Submit

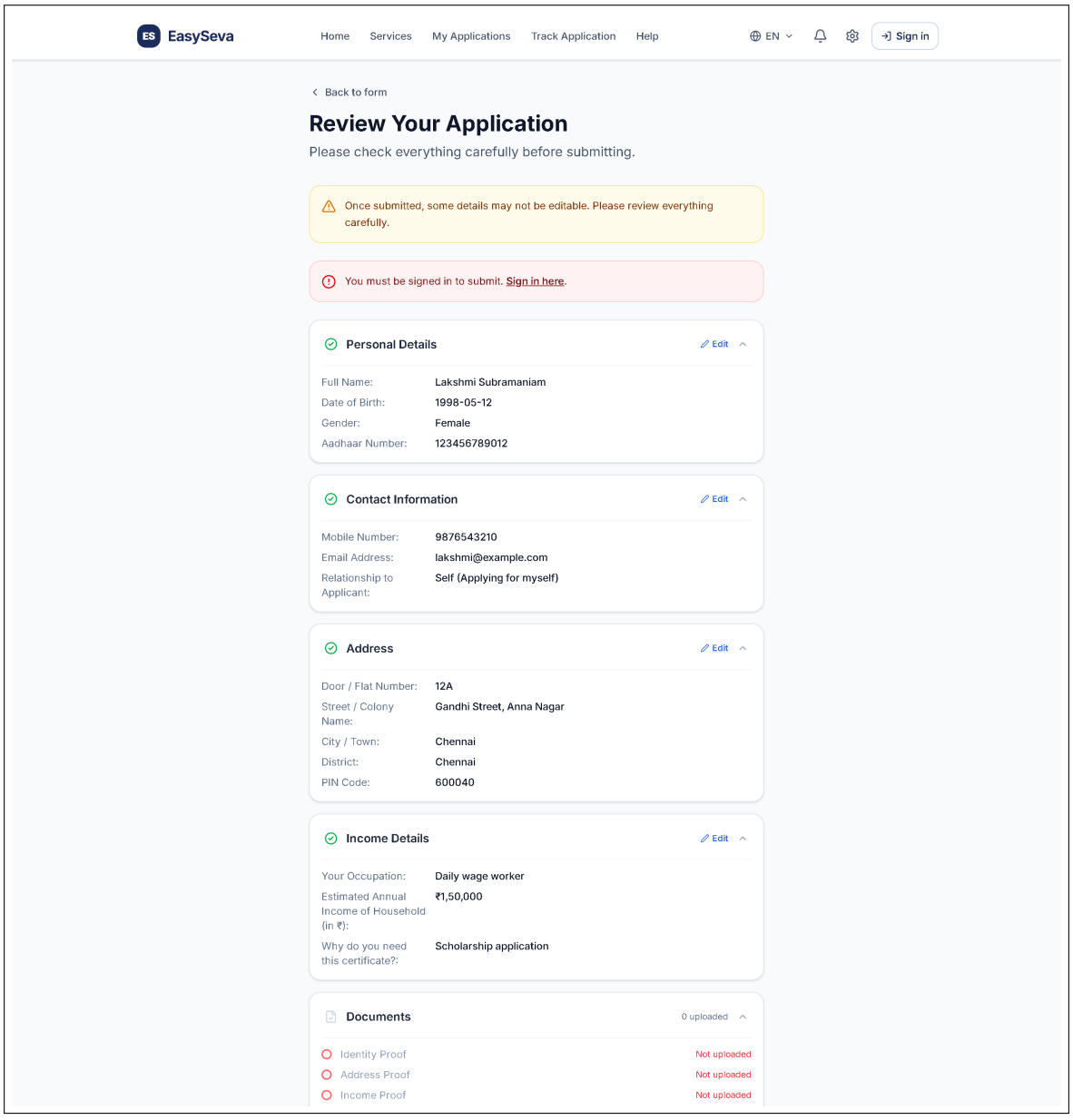


Figure 13. The review screen summarises every entered value and uploaded document in collapsible sections, each independently editable, with a final confirmation gate before submission.

Purpose. Let the user verify everything and correct anything before the irreversible submit.

- **Recognition rather than recall (N6).** *Summarise, don't quiz*

Every answer and document is shown back to the user, so they verify by recognition rather than trying to remember what they typed.

- **User control and freedom (N3, S6).** *Easy reversal*
  An "Edit" control on each section jumps straight back to that part of the form.
- **Error prevention (N5, S5).** *Confirm before commit*
  A warning banner and a confirmation dialog ("You cannot undo this action") guard the final submit.
- **Design dialogs to yield closure (S4).** *Clear end of task*
  The review marks a definite "check, then submit" close to the data-entry phase.

5.9 Confirmation

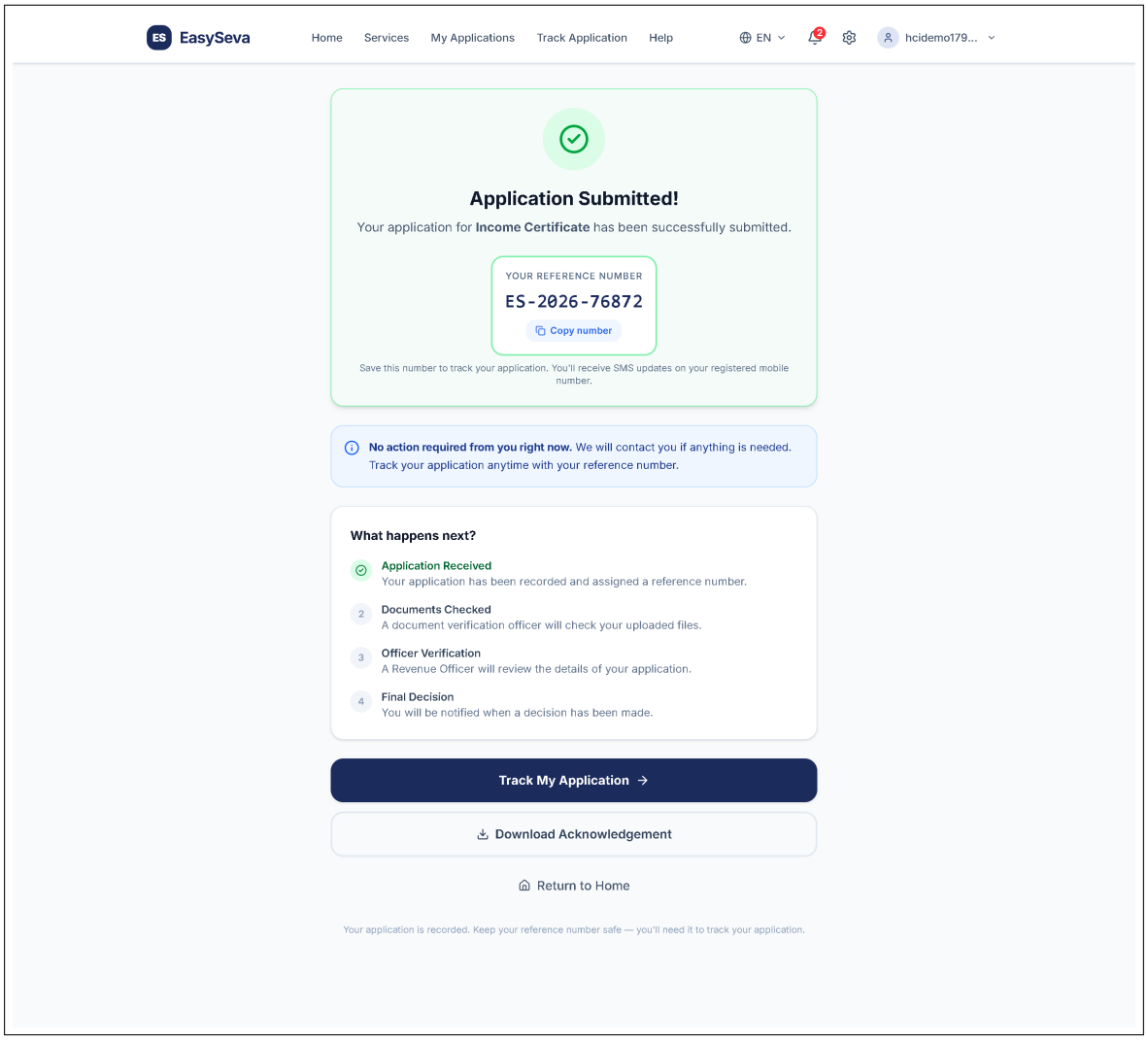


Figure 14. The confirmation screen acknowledges success, shows the reference number to save, and explains what happens next.

Purpose. Reassure the user that their application was received and tell them the next steps.

- **Visibility of system status (N1, S3).** *Unambiguous success*
  A success state plus a prominent reference number confirms the submission landed.

- **Design dialogs to yield closure (S4).** *Complete the task*
  The flow ends with a clear "done" state and a what-happens-next explanation, giving the user psychological closure.
- **Help users recover and plan (N10).** *Tell them what is next*
  Guidance on tracking and expected timelines prevents the common "what now?" anxiety.

5.10 Track an Application

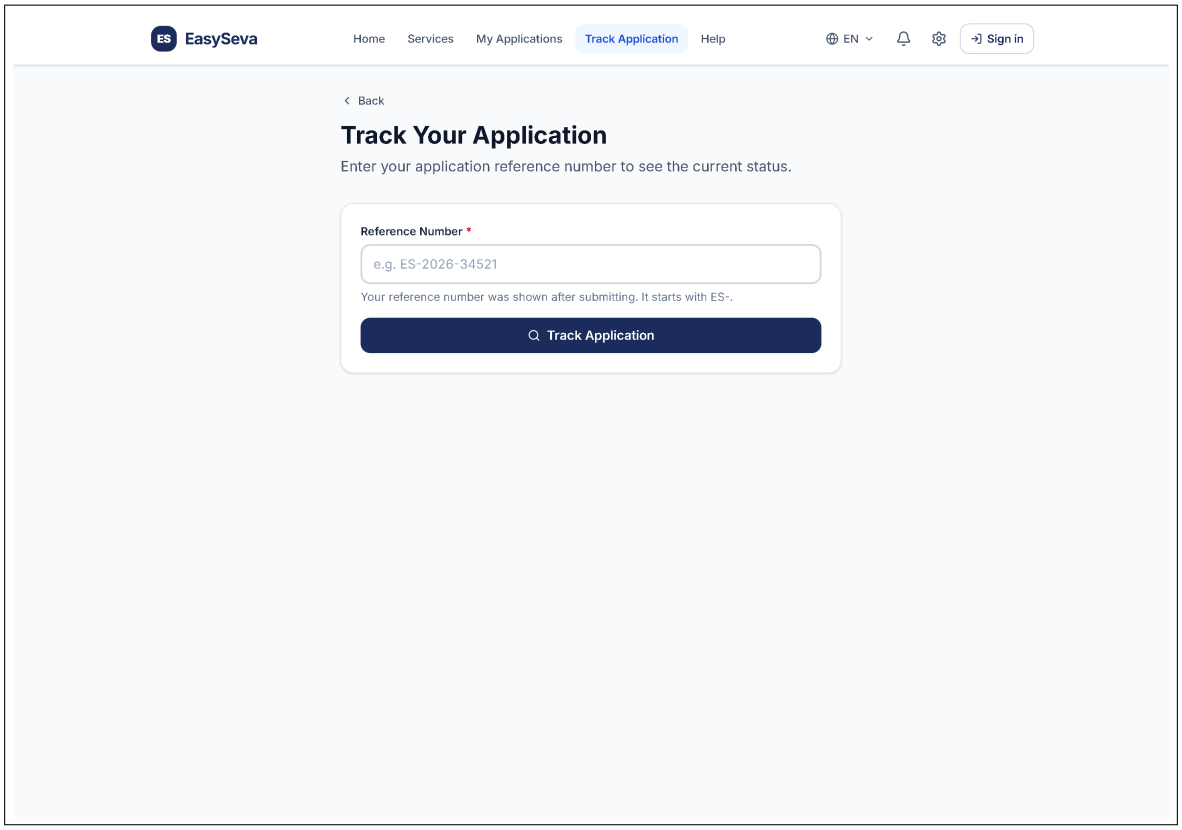


Figure 15. Tracking by reference number — available even without signing in — returns a plain-language status and timeline.

Purpose. Let anyone check progress quickly, including users who applied on someone's behalf.

- **Flexibility and efficiency (N7).** *Low-friction access*
  Tracking works without login, so a family member can check status with just the reference number.
- **Match with the real world (N2).** *Plain status*
  Statuses read as "Officer Verification" or "Documents Verified," not internal codes.
- **Visibility of system status (N1).** *Progress at a glance*
  A staged timeline shows exactly where the application is in the pipeline.

5.11 My Applications

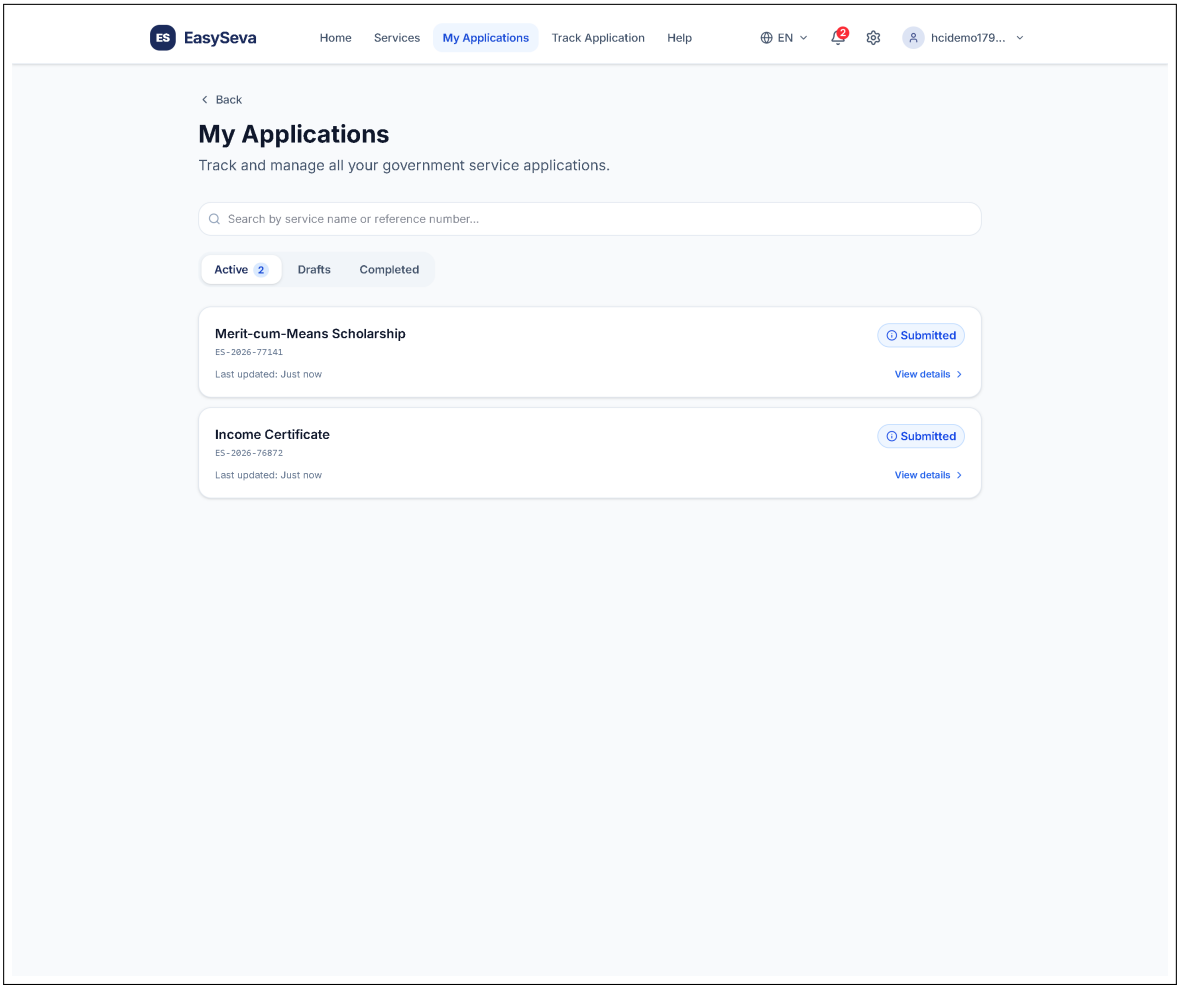


Figure 16. The applications list with Active / Drafts / Completed tabs, search, and a status badge on every item.

Purpose. A personal workspace to monitor and resume all applications.

- **Visibility of system status (N1, S3).** *Status everywhere*
  Each item carries a colour-coded status badge for an at-a-glance overview.
- **Flexibility and efficiency (N7).** *Filter and find*
  Tabs separate Active, Drafts, and Completed, and a search box finds items by name or reference.
- **Consistency and standards (N4, S1).** *Familiar patterns*
  The same card and badge components used elsewhere reappear here, so nothing must be relearned.

5.12 Notifications

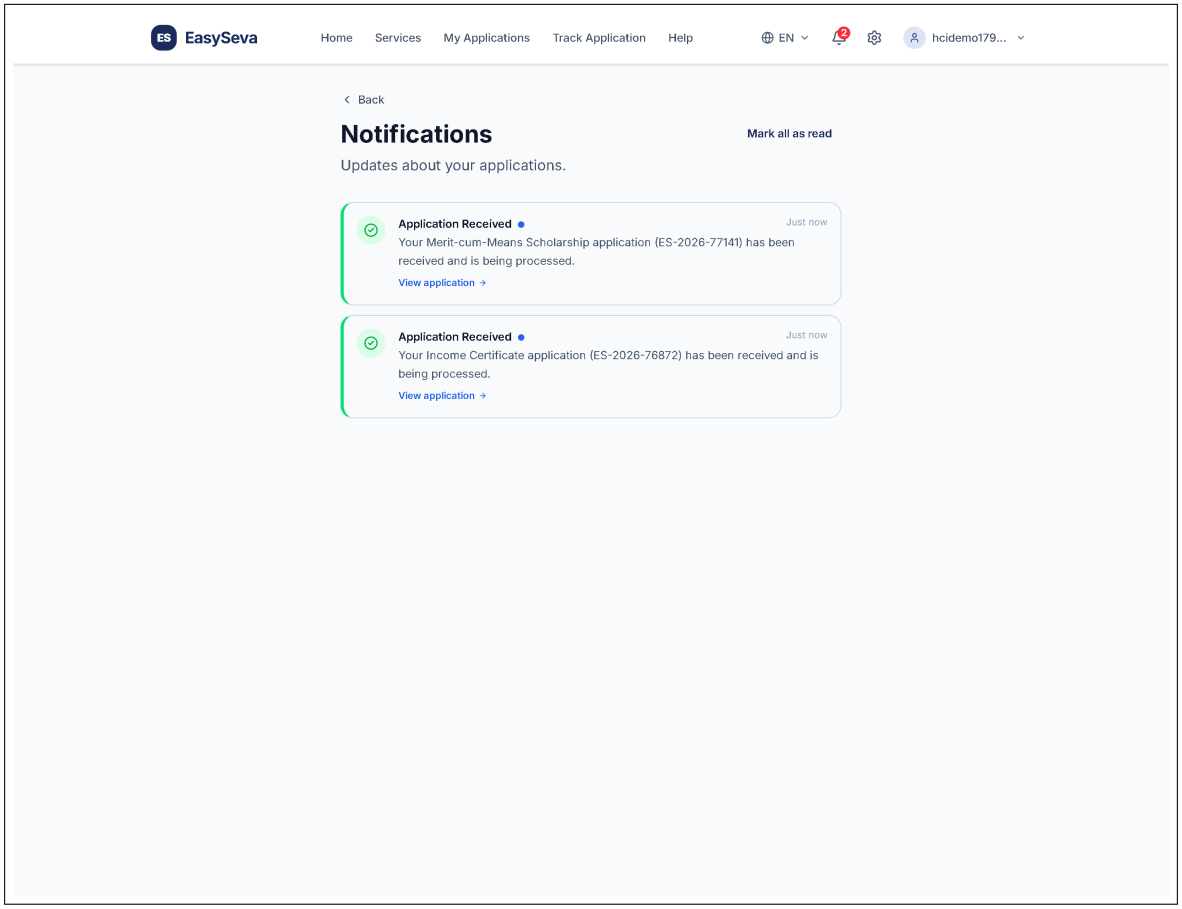


Figure 17. Notifications keep the citizen informed of each application events, with unread indicators and the ability to mark items read.

Purpose. Proactively surface changes so the user is never left guessing.

- **Visibility of system status (N1, S3).** *Proactive feedback*
  Status changes and required actions are pushed to the user rather than hidden.
- **User control and freedom (N3, S7).** *Manage the inbox*
  Users can mark a single item or all items as read, staying in control of their feed.
- **Recognition rather than recall (N6).** *Unread cues*
  An unread count on the header bell and per-item styling make new information obvious.

5.13 Profile

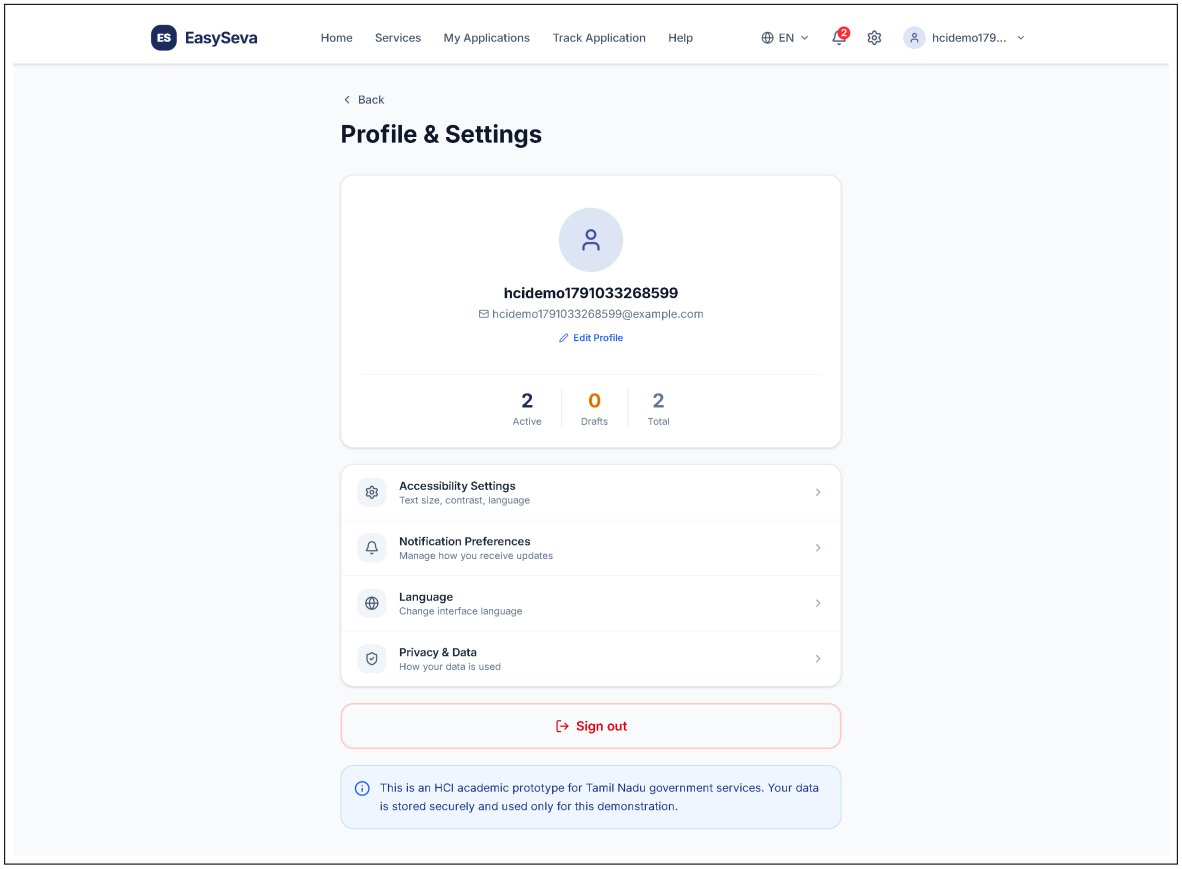


Figure 18. The profile page lets users view and edit their contact details with clear save feedback.

Purpose. Give users ownership of their own account information.

- **User control and freedom (N3, S7).** *Own your data*
  Users edit their name and phone number directly.
- **Informative feedback (S3, N1).** *Confirm the save*
  Saving produces an explicit "changes saved" confirmation.
- **Error prevention (N5).** *Validate contact details*
  Inputs are validated before they are accepted.

5.14 Sign In (OTP) and Verification

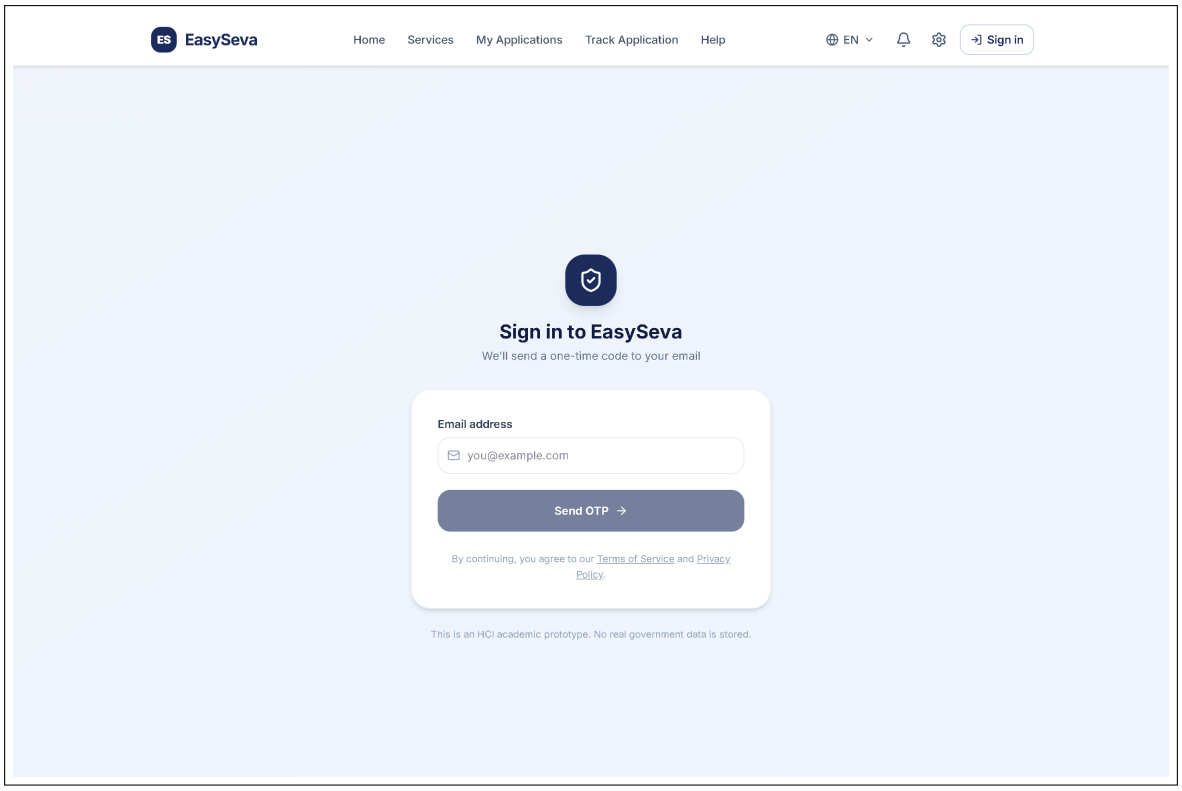


Figure 19. Passwordless sign-in: the user enters only an email address.

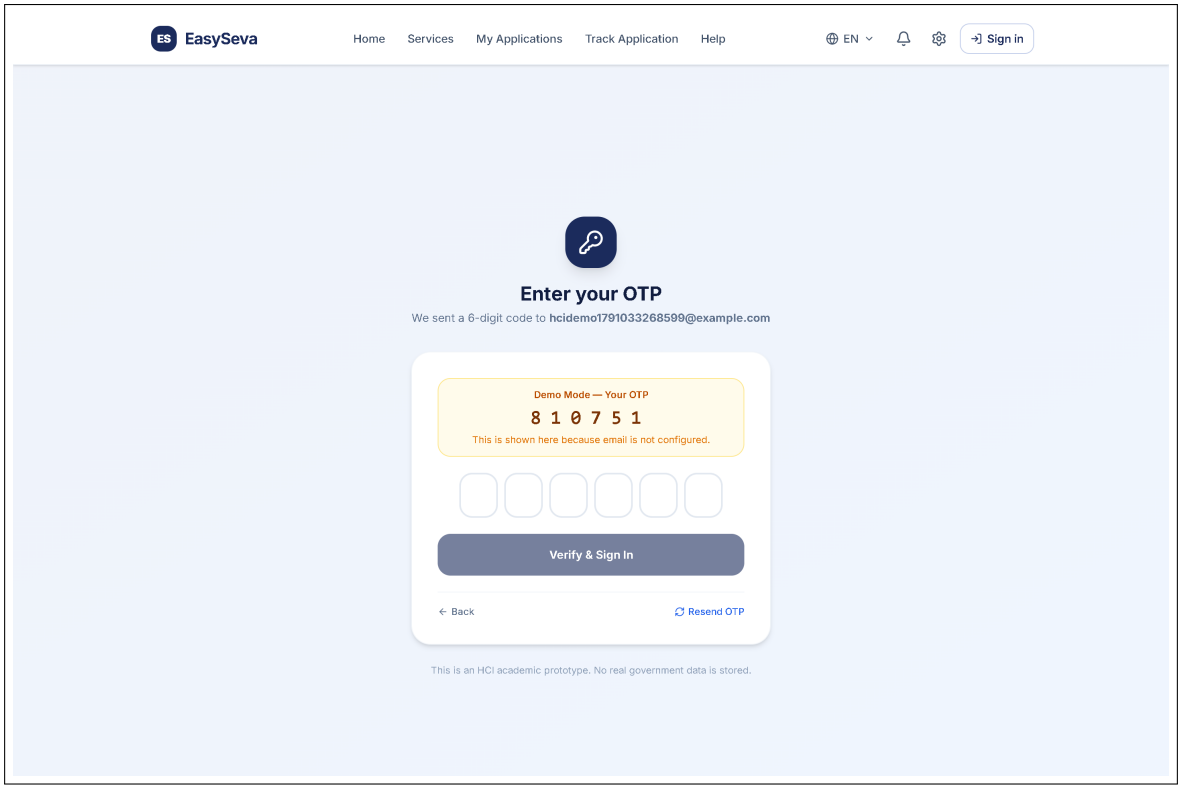


Figure 20. The verification screen with a six-box OTP input, a resend control with cooldown, and (in demo mode) the code shown on screen.

Purpose. Authenticate users with the least friction and the least chance of being locked out.

- **Error prevention (N5, S5).** *Remove a failure mode*
  A one-time code means there is no password to forget, reuse, or mistype.
- **Flexibility and efficiency (N7).** *Fast entry*
  The six-digit code can be pasted and auto-distributes across the boxes; focus advances automatically.
- **Visibility of system status (N1, S3).** *Show every state*
  "Sending OTP," "Verifying," and "Verified" states are shown explicitly.
- **Help users recover from errors (N9).** *Graceful recovery*
  A wrong code clears the boxes with a clear message; "Resend OTP" has a visible cooldown so users are not left stranded.
- **Affordance (D1).** *Obvious input*
  Six separate boxes afford a six-digit code and signal exactly how much to type.

5.15 Help Centre

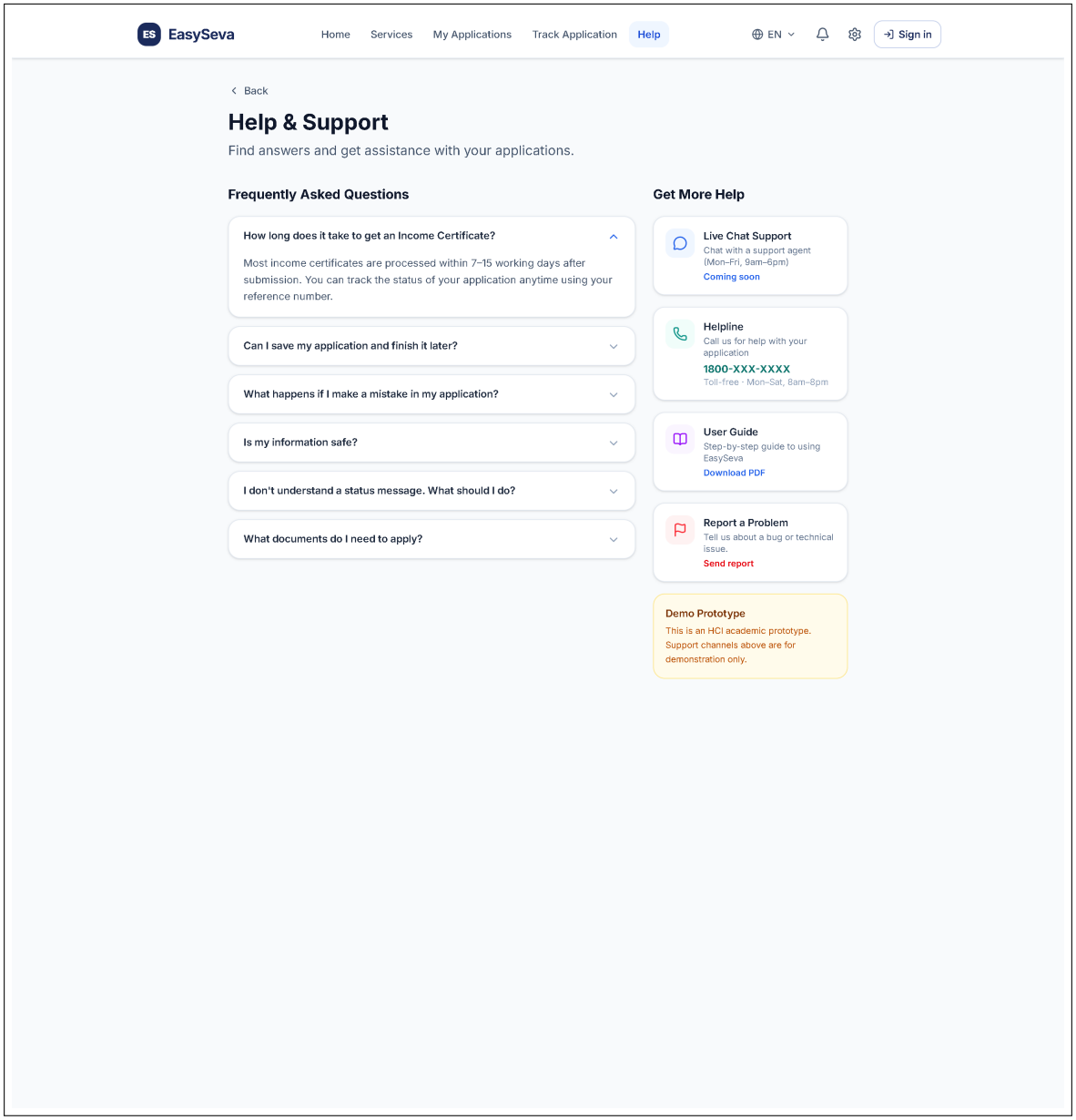


Figure 21. The Help page answers common questions in plain language.

Purpose. Provide searchable, task-oriented support.

- **Help and documentation (N10).** *Findable answers*
  Frequently asked questions are organised and phrased the way citizens ask them.
- **Match with the real world (N2).** *Plain answers*
  Answers avoid jargon and describe actions in user terms.

5.16 Responsive and Mobile Experience

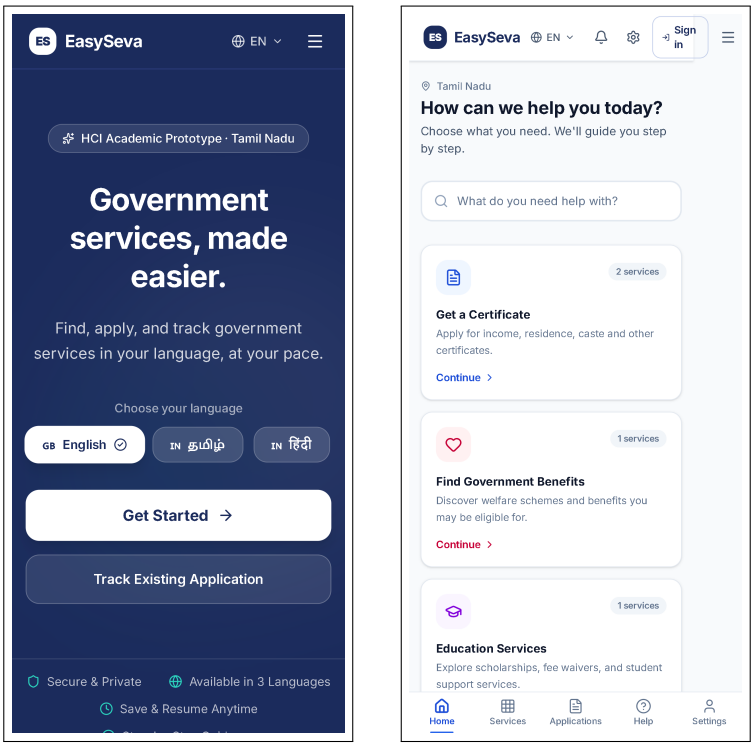


Figure 22. The landing page and home dashboard on a phone. The layout reflows to a single column and a bottom navigation bar appears for thumb-friendly reach.

Purpose. Serve the majority of citizens who access services on a phone.

- **Flexibility and efficiency (N7).** *Any device*
  Layouts reflow from multi- column (desktop) to single-column (mobile) without loss of function.
- **Cater to universal usability (S2).** *Reach and touch*
  A bottom navigation bar and large (44 px+) touch targets suit one-handed thumb use.
- **Consistency and standards (N4, S1).** *Same mental model*
  The mobile and desktop experiences share components and language, so skills transfer.

6 Cross-Cutting Principles

Some principles are not tied to a single screen but are woven throughout.

- **Help users recognise, diagnose, and recover from errors (N9).** *Plain, constructive errors*
  Error messages everywhere are human-readable and suggest a next step — e.g., "Please enter a valid 10-digit mobile number," or an OTP error that clears the field and invites a retry. Network failures fall back gracefully so the user is never stuck.
- **Keep users in control (S7, N3).** *Predictability*
  Actions are initiated by the user; destructive or irreversible actions (submit) require explicit confirmation and can be approached via a reversible draft.
- **Consistency (N4, S1, D5).** *One design language*
  Shared primitives and tokens make every button, card, badge, and field behave identically across the app.

- **Accessibility / POUR.** *Inclusive by default*
  Keyboard operability with visible focus, a skip link, ARIA roles and labels, scalable text, a high- contrast theme, and reduced-motion support make the app usable by a far wider range of people.
- **Conceptual model (D6).** *A clear mental picture*
  The consistent five-stage journey — discover, check, prepare, fill, confirm — gives users an accurate model of "how applying works," which transfers across all nine services.

7 Heuristic Coverage Matrix

Table 1 maps each Nielsen heuristic to representative places where it is applied in EasySeva.

Table 1. Where each usability heuristic is applied.

Heuristic	Representative use in EasySeva
Visibility of system status	Progress bars, step indicators, status badges and timeline, loading/saving/verifying states, confirmation with reference number.
Match with the real world	Task-based categories, plain-language labels and statuses, worked examples, rupee formatting.
User control and freedom	Back links and breadcrumbs, "Save & Finish Later," editable review, cancellable confirmation dialog, "Continue Anyway" on eligibility.
Consistency and standards	Shared Button/Card/Badge/Input primitives and design tokens across every page.
Error prevention	Eligibility pre-check, inline field validation, document type/size checks, confirmation before submit, OTP instead of passwords.
Recognition rather than recall	Document examples, review summary, breadcrumbs, visible service cards, unread cues.
Flexibility and efficiency	Three languages, search and guided finder, OTP paste, tracking without login, keyboard navigation.
Aesthetic and minimalist design	One primary action per screen, chunked catalogue and forms, uncluttered hero.
Recover from errors	Human-readable messages with next steps, OTP clear-and-retry, graceful network fallbacks.
Help and documentation	Service FAQs, Help centre, "Why are we asking this?" and "What can I use?" disclosures.

8 Conclusion

EasySeva demonstrates that the difference between an intimidating government portal and an empowering one is, largely, *applied HCI*. By grounding every screen in well-established principles — keeping status visible, speaking the user's language, preventing errors before they happen, favouring recognition over recall, keeping the user in control, and designing for universal access — the application turns a once-a-year, high-anxiety task into a calm, guided, and forgiving experience.

Crucially, these principles are not decoration: each one maps to a measurable reduction in the real failures citizens face — abandoned forms, rejected documents, lost progress, and

confusion over status. Good HCI, in other words, is what lets technology actually *serve* people. EasySeva is a working, deployed illustration of that idea.